

Wilcoxon Rank-Sum Test

Presentation Outline

1 Review: Two-Sample t -Test and CI Assumptions

2 Wilcoxon Rank-Sum Test

Two-Sample t -Test and CI Review

Compare value of quantitative variable between two populations.

Method - Hypothesis Test and Confidence Interval for $\mu_1 - \mu_2$

Based on Sample Means $\bar{Y}_1$ and $\bar{Y}_2$

- Sample Means are highly affected by outliers
- You must have values for all observations to calculate sample means.

This method may not be appropriate if you have outliers or other data issues.

Two-Sample t -Test and CI Assumptions

Population Variances are Equal ($\sigma_1^2 = \sigma_2^2$)

- Difficult assumption to test - rule of thumb: method is appropriate if ratio of s_1 to s_2 is between 0.5 and 2.
- If populations variances are very different, do we really want to just compare population means?

Sampling distribution of $\bar{Y}_1 - \bar{Y}_2$ is a Normal distribution

- One or both sample sizes may not be large enough for this assumption to be true.

This method may not be appropriate if assumptions above are not met by your data.

Wilcoxon Rank-Sum Test

Compares values of quantitative variable between two groups

Resistant to outliers

Only requires ranking observations, so can be used in situations where some observations are observed in a range of values.

Several ways to set up hypothesis test. We will use:

- H_0 : Distribution of quantitative variable is the same in the two populations.
- H_a : Distribution of quantitative variable is different between the two populations.

Wilcoxon Rank-Sum Test Statistic

Steps to calculating the Wilcoxon Rank-Sum Test Statistic

- ➊ Combine the data from the two groups into a single data set
- ➋ Order the $n_1 + n_2$ observations from smallest to largest
- ➌ Assign ranks
 - The smallest observation gets rank=1, the second smallest gets rank=2, etc...
 - For tied observations, average the ranks
- ➍ Define $W = \text{sum of ranks for group 1} - \frac{n_1(n_1+1)}{2}$

Wilcoxon Rank-Sum Test Statistic

Assuming the null hypothesis of equivalent distributions is true, the expected value and variance of W is:

$$E_0(W) = \frac{n_1 * n_2}{2}$$

$$V_0(W) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

Wilcoxon Rank-Sum Test Statistic and p -value

Test Statistic:

For large sample sizes n_1 and n_2 , the sampling distribution of:

$$z = \frac{|W - E_0(W)| - 0.5}{\sqrt{V_0(W)}}$$

is well-approximated by the standard Normal distribution.

Approximate p -value:

$$2 * P(Z > |z|)$$

Example - Wilcoxon Rank-Sum Test

Creativity Data Set

- Group 1 = Extrinsic
 - Sum of Ranks = 426
- Group 2 = Intrinsic
 - Sum of Ranks = 702

Test Statistic:

$$\begin{aligned} W &= \text{sum of ranks for Extrinsic group} - \frac{n_1(n_1 + 1)}{2} \\ &= 426 - \frac{23 * 24}{2} = 150 \end{aligned}$$

Example - Wilcoxon Rank-Sum Test

$$E_0(W) = \frac{n_1 * n_2}{2} = \frac{23 * 24}{2} = 276$$

$$V_0(W) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12} = \frac{23 * 24 (23 + 24 + 1)}{12} = 2208$$

$$z = \frac{|W - E_0(W)| - 0.5}{\sqrt{V_0(W)}} = \frac{|150 - 276| - 0.5}{\sqrt{2208}} = 2.67$$

Example - Wilcoxon Rank-Sum Test

p -value:

$$2 * P(Z > 2.67) = 0.0076$$

Conclusion:

We have very strong evidence the distribution of creativity scores is different between the Extrinsic and Intrinsic treatment groups.